

[GETNET] - Recuperação de Standby Database via RMAN

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1. Introdução

Durante a análise do ambiente Data Guard entre os bancos **STRANAPPHMLDCS (Primary)** e **STRANAPPHMLDCN (Standby)**, foi identificado um atraso de replicação (apply e transport lag superior a 150 dias).

Apesar da divergência, optou-se **por não realizar a reconstrução completa do Data Guard**, visto que os **parâmetros de configuração e conexões entre os hosts estavam íntegros** — as entradas de **tnsnames.ora**, **listener.ora**, e as configurações de **DG_BROKER** estavam válidas, e o **canal de comunicação entre os bancos (tnsping)** respondia corretamente.

A abordagem adotada visou **preservar a estrutura existente e realinhar o standby** através do **RMAN DUPLICATE / RECOVER FROM SERVICE**, garantindo consistência sem necessidade de reinstalação completa.

2. Objetivo

Reestabelecer a sincronização do banco **standby STRANAPPHMLDCN** com o **primary STRANAPPHMLDCS**, mantendo a configuração atual do Data Guard e restabelecendo o processo de Managed Recovery.

3. Acesso ao Data Guard Broker

Conectar ao **DGMGRL** e verificar o status da configuração:

```
dgmgrl /
show configuration;
show database STRANAPPHMLDCN;
```

Objetivo: identificar lag de transporte e aplicar, além de checar erros de sincronização.

```
[oracle@GNCASSHL09252 ~]$ dgmgrl /
DGMGRL for Linux: Release 19.0.0.0 - Production on Wed Oct 8 18:04:55 20
25
Version 19.26.0.0.0
Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights reserv
ed.
Welcome to DGMGRL, type "help" for information.
Connected to "STRANAPPHMLDCS"
Connected as SYSDBA.
DGMGRL> show configuration;

Configuration - DR_STRANAPPHML

Protection Mode: MaxPerformance
Members:
STRANAPPHMLDCS - Primary database
Error: ORA-16724: cannot resolve gap for one or more members
STRANAPPHMLDCN - Physical standby database
Warning: ORA-16809: multiple warnings detected for the member

Fast-Start Failover: Disabled

Configuration Status:
ERROR (status updated 43 seconds ago)

DGMGRL> DGMGRL> [oracle@GNCASSHL09252 ~]$ tnsping STRANAPPHMLDCN
TNS Ping Utility for Linux: Version 19.0.0.0.0 - Production on 08-OCT-2025 18:08:44
Copyright (c) 1997, 2024, Oracle. All rights reserved.

Used parameter files:
/u01/app/oracle/product/19.23/dbhome_1/network/admin/sqlnet.ora

Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCP)(HOST = GNCASSHL09253)(PORT = 1521)) (CONNECT_DATA = (SERVER = DEDICATED) (SERVICE_NAME = STRANAPPHMLDCN)))
OK (10 msec)
```

4. Colocar o Standby em Estado de MOUNT

No servidor **Standby**:

```
startup force mount;
```

5. Desabilitar o Apply no Data Guard Broker

No **Primary**, ajustar o estado do standby:

```
dgmgrl /
edit database STRANAPPHMLDCN set state='APPLY-OFF';
```

Objetivo: interromper o processo de recuperação gerenciado antes de iniciar a sincronização manual.

```
[oracle@GNCASSHL09252 ~]$ dgmgrl /
DGMGRL for Linux: Release 19.0.0.0.0 - Production on Wed Oct 8 18:09:23 2025
Version 19.26.0.0.0

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Welcome to DGMGRL, type "help" for information.
Connected to "STRANAPPHMLDCS"
Connected as SYSDBA.
DGMGRL> show database STRANAPPHMLDCN

Database - STRANAPPHMLDCN

Role:                PHYSICAL STANDBY
Intended State:       APPLY-ON
Transport Lag:        154 days 13 hours 24 minutes 46 seconds (computed 0 seconds ago)
Apply Lag:            154 days 13 hours 34 minutes 26 seconds (computed 0 seconds ago)
Average Apply Rate:   0 Byte/s
Real Time Query:      OFF
Instance(s):
  STRANAPPHML

Database Warning(s):
  ORA-16853: apply lag has exceeded specified threshold
  ORA-16855: transport lag has exceeded specified threshold

Database Status:
WARNING

DGMGRL> edit database STRANAPPHMLDCN set state='APPLY-OFF';
Succeeded.
```

6. Cancelar o Managed Recovery

No Standby:

```
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
```

Caso o erro `ORA-16136: Managed Standby Recovery not active` ocorra, ignore — indica que o apply já estava parado.

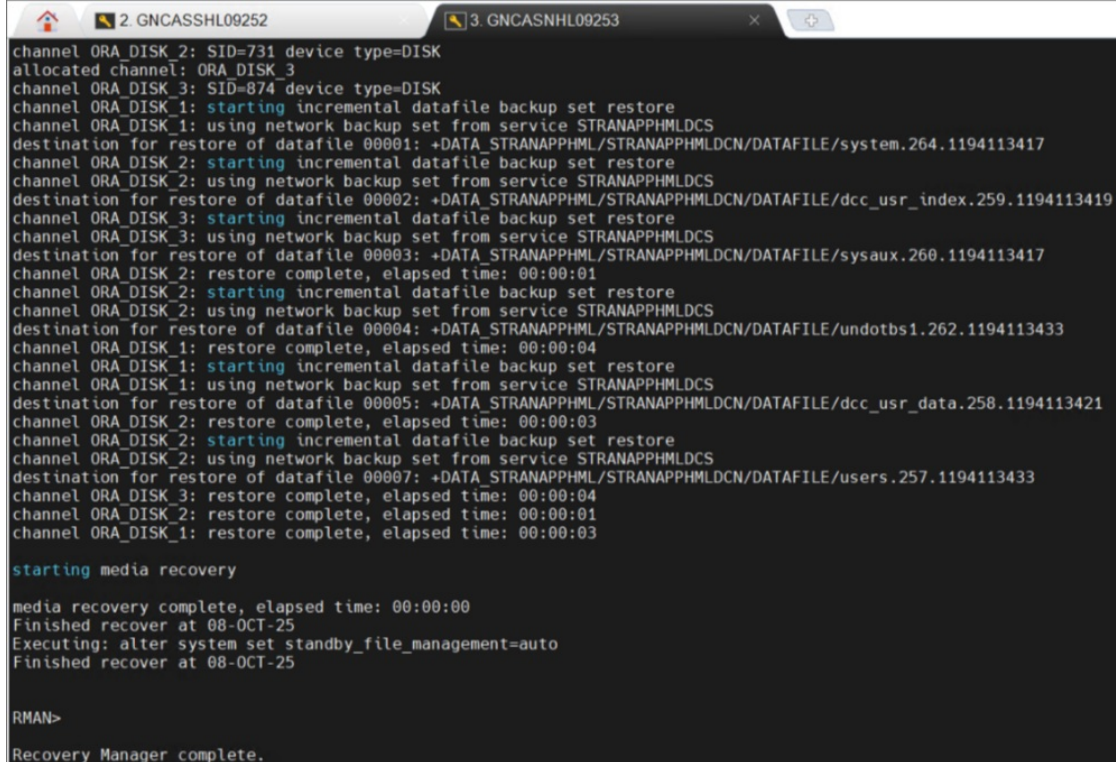
7. Recuperar o Standby via RMAN

No Standby, executar:

```
rman target /
```

```
recover standby database from service STRANAPPHMLDCS;
```

Objetivo: aplicar os logs de redo recebidos diretamente do primary, sincronizando os arquivos de dados.



```
channel ORA_DISK_2: SID=731 device type=DISK
allocated channel: ORA_DISK_3
channel ORA_DISK_3: SID=874 device type=DISK
channel ORA_DISK_1: starting incremental datafile backup set restore
channel ORA_DISK_1: using network backup set from service STRANAPPHMLDCS
destination for restore of datafile 00001: +DATA_STRANAPPHML/STRANAPPHMLDCN/DATAFILE/system.264.1194113417
channel ORA_DISK_2: starting incremental datafile backup set restore
channel ORA_DISK_2: using network backup set from service STRANAPPHMLDCS
destination for restore of datafile 00002: +DATA_STRANAPPHML/STRANAPPHMLDCN/DATAFILE/dcc_usr_index.259.1194113419
channel ORA_DISK_3: starting incremental datafile backup set restore
channel ORA_DISK_3: using network backup set from service STRANAPPHMLDCS
destination for restore of datafile 00003: +DATA_STRANAPPHML/STRANAPPHMLDCN/DATAFILE/sysaux.260.1194113417
channel ORA_DISK_2: restore complete, elapsed time: 00:00:01
channel ORA_DISK_2: starting incremental datafile backup set restore
channel ORA_DISK_2: using network backup set from service STRANAPPHMLDCS
destination for restore of datafile 00004: +DATA_STRANAPPHML/STRANAPPHMLDCN/DATAFILE/undotbs1.262.1194113433
channel ORA_DISK_1: restore complete, elapsed time: 00:00:04
channel ORA_DISK_1: starting incremental datafile backup set restore
channel ORA_DISK_1: using network backup set from service STRANAPPHMLDCS
destination for restore of datafile 00005: +DATA_STRANAPPHML/STRANAPPHMLDCN/DATAFILE/dcc_usr_data.258.1194113421
channel ORA_DISK_2: restore complete, elapsed time: 00:00:03
channel ORA_DISK_2: starting incremental datafile backup set restore
channel ORA_DISK_2: using network backup set from service STRANAPPHMLDCS
destination for restore of datafile 00007: +DATA_STRANAPPHML/STRANAPPHMLDCN/DATAFILE/users.257.1194113433
channel ORA_DISK_3: restore complete, elapsed time: 00:00:04
channel ORA_DISK_2: restore complete, elapsed time: 00:00:01
channel ORA_DISK_1: restore complete, elapsed time: 00:00:03

starting media recovery

media recovery complete, elapsed time: 00:00:00
Finished recover at 08-OCT-25
Executing: alter system set standby_file_management=auto
Finished recover at 08-OCT-25

RMAN>

Recovery Manager complete.
```

8. Reativar o Apply no Broker

Após a recuperação bem-sucedida:

dgmgri /

edit database STRANAPPHMLDCN set state='APPLY-ON';

Objetivo: retomar o processo de **Managed Standby Recovery** controlado pelo Data Guard Broker.

```
DGMGRL> edit database STRANAPPHMLDCN set state='APPLY-ON';
Succeeded.
```

9. Abrir e Validar o Banco

No Standby:

ALTER DATABASE OPEN;

```
col HOST_NAME HEADING 'HOSTNAME' format a15;
SELECT DBID, NAME, DB_UNIQUE_NAME, INSTANCE_NAME, CREATED, LOG_MODE, DATABASE_ROLE, OPEN_MODE, IT.STATUS, IT.HOST_NAME, DT.switchover_status
FROM gv$DATABASE DT INNER JOIN gv$instance IT ON DT.INST_ID = IT.INST_ID;
SQL> SQL> SQL> SQL> SQL> SQL>
SELECT PROCESS, STATUS FROM V$MANAGED_STANDBY;

DBID NAME DB_UNIQUE_NAME INSTANCE_NAME CREATED LOG_MODE DATABASE_ROLE OPEN_MODE STATUS HOSTNAME
-----
866835326 STRANAPP STRANAPPHMLDCN STRANAPPHML 24-JUN-24 ARCHIVELOG PHYSICAL STANDBY READ ONLY WITH APPLY OPEN GNCASNH
ALLOWED

SQL> SQL>
PROCESS STATUS
-----
DGRD ALLOCATED
ARCH CONNECTED
DGRD ALLOCATED
ARCH CONNECTED
ARCH CONNECTED
ARCH CONNECTED
ARCH CLOSING
RFS IDLE
RFS IDLE
MRP0 APPLYING_LOG

10 rows selected.

SQL> SELECT ARCH.THREAD# "Thread", ARCH.SEQUENCE# "LLGPD Sequence Received", APPL.SEQUENCE# "LLGPD Sequence Applied", (ARCH.SEQUENCE# - APPL
Difference"
FROM (SELECT THREAD#, SEQUENCE# FROM V$ARCHIVED_LOG WHERE (THREAD#, FIRST_TIME) IN (SELECT THREAD#, MAX(FIRST_TIME) FROM V$ARCHIVED_LOG GROUP
ARCH,
2 3 (SELECT THREAD#, SEQUENCE# FROM V$LOG_HISTORY WHERE (THREAD#, FIRST_TIME) IN (SELECT THREAD#, MAX(FIRST_TIME) FROM V$LOG_HISTORY GR
#)) APPL
4 WHERE ARCH.THREAD# = APPL.THREAD# ORDER BY 1;

Thread LLGPD Sequence Received LLGPD Sequence Applied Difference
-----
1 51065 51065 0

SQL>
```

Depois, validar no Primary:

dgmgri /

show configuration;

Objetivo: garantir que o status geral da configuração retorne a **SUCCESS**.

10. Conclusão

Com o procedimento, o banco standby **STRANAPPHMLDCN** foi sincronizado novamente com o banco **primary STRANAPPHMLDCS** sem necessidade de reconstrução total. A solução adotada reduziu tempo de indisponibilidade e manteve a integridade da configuração Data Guard existente.